

Snap Fit Four Housings SOP (1x Episode: 4 Housings)

One episode snaps together all four housings. The episode ends when the fourth closed housing sits in the finished tray.

Each housing is two parts: a **bottom half** and a **top half** that clips onto it. The table begins with four bottom halves in the **bottom tray**, four top halves on the **top stand** at the right, an empty **assembly nest** just right of the center of the table, and an empty **finished tray** with four slots just left of center.

The order never changes for each housing: set the bottom half into the nest, lower the top half straight down onto it, click the four clips in the order **1 back** → **2 front** → **3 right** → **4 left**, press all four edges home, look at the seam all the way round, then set the closed housing in the next slot of the finished tray. One housing is finished and trayed before the next bottom half is picked up.

The right gripper does the work: it carries every top half and every finished housing, aligns the halves, presses each clip to a click, presses the edges home, and trays the finished housing. The left gripper supports: it presses the top half down on its **hold pad** and holds the halves together through the clip pass and the edge press, and it steadies the finished tray while a housing goes into a slot.

The table is set up in one of three ways. Only the bottom tray moves; the top stand, the assembly nest, and the finished tray are in the same place in all three.

- **Config L:** the bottom tray is at the back-left.
- **Config M:** the bottom tray is at the front-center, in front of the finished tray and the nest.
- **Config R:** the bottom tray is at the front-right.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the bottom tray starts at the back-left (**Config L**), the front-center (**Config M**) or the front-right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the gripper on the bottom tray's side takes each bottom half and sets it in the nest — the left gripper in Config L and M, the right gripper in Config R. No arm reaches across the table.
- **Fixed roles:** the right gripper carries every top half, clicks the clips, presses the edges, and trays every housing; the left gripper holds the hold pad and steadies the finished tray. The

build order and the slot order do not change with the config.

Setup

Complete both checklists before recording starts.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the assembly nest, the bottom tray, the top stand, and all four slots of the finished tray.
3. The nest is visible from above and from the front, so the seam between the two halves can be seen once the top half is on.
4. Both arms start at home with both grippers open.
5. The table is clear of everything but the nest, the two supply spots, and the finished tray.
6. Each arm can reach its own work without stretching: the right gripper reaches the bottom tray (Config R), the top stand, the nest, and all four slots of the finished tray; the left gripper reaches the bottom tray (Config L and M), the top of the nest, and the left edge of the finished tray.

Materials checklist

1. One **assembly nest** sits just right of the center of the table, square to the front edge, empty. It does not move when a gripper presses down in it.
2. Four **bottom halves** stand open side up in the **bottom tray**, each with its **front mark** pointing toward the front edge. The tray is in the start zone for this episode's config, and the other two zones are empty:
 - **Config L:** back-left
 - **Config M:** front-center, in front of the finished tray and the nest
 - **Config R:** front-right
3. Four **top halves** sit on the **top stand** at the right, open side down, each with its **front mark** pointing toward the front edge and its **hold pad** facing up.
4. Every bottom half has four clean, open **catch windows**, one on each of its four outside walls.
5. Every top half has four unbroken **clips**, one inside each of its four walls, and four clean **press pads** on its top face.

6. One **finished tray** lies flat just left of center with four empty **slots** in a row: **slot 1** nearest the front edge, then slots 2, 3, and 4 going back.
7. No half is cracked, warped, or bent, and no clip is snapped off or bent flat.
8. Keep the left side of the table clear apart from the bottom tray in Config L. The left gripper comes in from there onto the nest and the finished tray.
9. Before collection, confirm by hand that each top half drops onto each bottom half without being forced, that all four clips click, that the closed housing lifts out of the nest in one piece, and that a closed housing drops into a slot and sits flat.

Workspace layout

- **Assembly spot:** just right of the center of the table — the assembly nest, where every housing is built
- **Bottom tray:** four bottom halves — back-left (Config L), front-center (Config M), or front-right (Config R)
- **Right:** the top stand with four top halves
- **Just left of center:** the finished tray, slots 1 to 4 running front to back
- **Left side:** kept clear apart from the bottom tray in Config L, so the left gripper can come in onto the nest and the finished tray

Arm assignments

- **Left gripper:** presses the top half down on its **hold pad** and holds the two halves together through the clip pass and the edge press, then steadies the finished tray while the right gripper sets a housing into its slot. In Config L and M it also carries each bottom half from the bottom tray into the nest.
- **Right gripper:** carries each bottom half into the nest in Config R, carries each top half and lowers it on, presses each of the four clips to a click, presses the four edges home, and carries each finished housing to its slot in the finished tray.

Vocabulary

- **Housing:** one bottom half and one top half clipped together. Four housings are built in the episode.
- **Bottom half:** the open-top shell that goes into the nest. Its four outside walls carry the catch windows.
- **Bottom tray:** where the four bottom halves start — back-left (**Config L**), front-center (**Config M**), or front-right (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Top half:** the shell that clips onto the bottom half. Its four inside walls carry the clips, and its top face carries the hold pad and the four press pads.

- **Assembly nest:** the shallow holder just right of center that holds one bottom half square and still while the top half is clipped on. Only one housing is in the nest at a time.
- **Front mark:** the painted mark on one wall of each half. It points toward the **front edge** when the half is the right way round. The two marks line up when the halves are the right way round on each other.
- **Clip:** the small hook inside one wall of the top half. It snaps into the catch window on the matching wall of the bottom half.
- **Catch window:** the square hole in a bottom half wall that one clip snaps into.
- **Latched:** the hook of a clip shows through its catch window and fills it. All four clips latched means the housing is closed.
- **Click:** the single sound and give when a clip drops into its catch window. It is the signal to stop pressing that pad.
- **Press pad:** one of the four flat pads on the top face of the top half, each one over its own clip. The right gripper presses straight down on a pad to click that clip.
- **Pad numbers:** pad **1** is at the back, **2** at the front, **3** at the right, and **4** at the left. Clips are clicked in the order 1, 2, 3, 4.
- **Hold pad:** the flat pad in the middle of the top face. It is the only place the left gripper presses.
- **Rest position:** the top half sits down on the bottom half with the walls lined up all the way round, but no clip has clicked yet. The top half stays there on its own when the right gripper opens.
- **Seam:** the joint line running all the way round the housing where the two halves meet.
- **Daylight:** a gap in the seam you can see through or see a shadow line in. A closed seam shows none.
- **Flush:** the outside wall of the top half lines up with the outside wall of the bottom half, with no step between them.
- **Edge press:** the closing pass over all four edges of the top face after the four clips have clicked.
- **Finished tray:** the tray just left of center that holds the closed housings. It stays flat on the table and is never lifted.
- **Slot numbers:** slot **1** is nearest the front edge, then **2**, **3**, and **4** going back. Housings fill the slots in that order.
- **Seated flat:** the part sits down on the bottom of the nest or the slot, level, and does not rock when a gripper touches it.

Steps

Run Steps 1–6 on one housing at a time: housing 1 first, then 2, then 3, then 4. Finish and tray a housing before picking up the next bottom half. End the episode with Step 7.

Step 1 depends on where the bottom tray is: in Config L and M the **left gripper** takes each bottom half and sets it in the nest; in Config R the **right gripper** does. Every other line in Steps 1–7 is the same in all three configs.

Step 1: Set a bottom half into the nest

Goal: one bottom half sits flat in the nest, open side up, front mark toward the front edge.

Look where the bottom tray is before reaching for the first bottom half.

- **IF the bottom tray is at the back-left (Config L):** with the **left gripper**, pinch the next **bottom half** across its left and right outside walls, lift it out of the tray, and carry it forward and right to the nest.
- **IF the bottom tray is at the front-center (Config M):** with the **left gripper**, pinch the next **bottom half** across its left and right outside walls, lift it out of the tray, and carry it straight back to the nest.
- **IF the bottom tray is at the front-right (Config R):** with the **right gripper**, pinch the next **bottom half** across its left and right outside walls, lift it out of the tray, and carry it back and left to the nest.

Then, in all three:

- Carry it in one go, straight from the bottom tray to the nest. Do not set it down on the table on the way.
- Hold it level, open side up, with its **front mark** pointing toward the **front edge**.
- Lower it straight down into the nest until it is **seated flat**, then open the gripper and lift clear.
- Carry one bottom half at a time. Do not lift two parts together.

Expected state: one bottom half sits in the nest, open side up, and the top stand is untouched.

Check: the bottom half is seated flat in the nest, its front mark points toward the front edge, and all four **catch windows** are clear and open. If it rocks, sits high on one side, or its mark points the wrong way, lift it straight out with the same gripper and set it in again.

Step 2: Lower the top half onto the bottom half

Goal: the top half rests on the bottom half in **rest position**, the right way round, walls lined up all the way round.

- With the **right gripper**, pinch the next **top half** across its left and right outside walls and lift it off the top stand.
- Carry it in one go, straight from the top stand to the nest. Do not set it down on the table on the way.
- Hold it level, open side down, with its **front mark** pointing toward the **front edge**, so the two front marks line up.

- With the **right gripper**, lower it straight down onto the bottom half until its walls sit outside the bottom half's walls all the way round and it stops.
- Lower it level. Do not land it on one corner, and do not slide, drag, or twist it on the bottom half to line the walls up.
- If it lands crooked or hangs up on a wall, lift it straight up with the **right gripper**, hold it level again, and lower it once more.
- Open the **right gripper** and lift clear only once the top half stays in **rest position** on its own.

Expected state: the top half rests on the bottom half, both front marks point toward the front edge, no clip has clicked, and the top stand holds one fewer top half.

Check: the two front marks line up, the walls line up all the way round with no corner hanging over, and the top half sits level and does not rock. If it sits high on one side or hangs over a wall, lift it straight up with the right gripper and lower it on again.

Step 3: Hold the halves and click the four clips

Goal: all four clips are latched, each clicked once, in the order **1 back** → **2 front** → **3 right** → **4 left**.

- With the **left gripper**, press down on the **hold pad** in the middle of the top face and hold the halves together for the whole of this step.
- With the **right gripper**, press the pads in **pad order 1, 2, 3, 4**. Press each pad straight down.
- Give each clip one click. Do not go back to a pad that has already clicked.

3.1 Click one clip

- With the **right gripper**, close the gripper and set its tip down flat on the current **press pad**.
- Press straight down in one steady move until the clip **clicks** once.
- Stop pressing the moment it clicks, and lift the gripper straight up off the pad.
- Press down only. Do not rock the top half, do not press at an angle, and do not push the housing sideways in the nest.

3.2 Work round the four pads

- Repeat 3.1 on the next pad in pad order, until all four clips have clicked.
- If a pad presses with no click, lift the **right gripper** straight up and press that pad once more. If it still does not click, go on to the next pad and log the housing at Step 5.
- Keep the **left gripper** on the hold pad until all four pads have been pressed.

Expected state: all four clips are latched, the top half sits down on the bottom half all the way round, and the left gripper is still on the hold pad.

Check: each of the four **catch windows** shows a clip hook filling it, and the top half no longer lifts when the right gripper touches it. If a window is empty, press that pad again with the right gripper.

Step 4: Press all four edges home

Goal: the seam is closed all the way round, with the walls **flush** on all four sides.

- Keep the **left gripper** pressing the **hold pad** for the whole of this step.
- With the **right gripper**, press straight down on the top face along the **back edge**, midway between the corners, and hold for 2 seconds.
- With the **right gripper**, press the same way along the **front edge**, then the **right edge**, then the **left edge**, holding each for 2 seconds.
- Press each edge straight down. Do not press the corners, do not slide the gripper along the face, and do not pry under the seam with the gripper tip.
- With the **left gripper**, let go of the hold pad and move clear of the nest once all four edges are pressed.

Expected state: the housing sits closed in the nest, both grippers are clear of it, and the seam is in plain view.

Check: the top half does not rock when the right gripper presses on it, and no edge stands higher than the others. If an edge stands high, press that edge again with the right gripper.

Step 5: Look at the seam all the way round

Goal: the seam is confirmed closed on all four sides before the housing is trayed.

- Keep both grippers clear of the housing while looking at it.
- Look at the seam on the **front** side, then the **right**, then the **back**, then the **left**.
- On each side, look for **daylight** in the seam, for a wall that is not **flush**, and for a **catch window** with no clip hook in it.
- If one side shows daylight or a step, press that edge again with the **right gripper**, straight down, held for 2 seconds, then look at that side once more.
- If that side still shows daylight, or a catch window is still empty, go on to Step 6 with the housing as it is and log the episode for a station check.

Expected state: the closed housing sits in the nest, checked on all four sides, with both grippers clear.

Step 6: Tray the finished housing

Goal: the closed housing sits flat in the next empty slot of the finished tray, front mark toward the front edge.

- With the **left gripper**, press the left edge of the **finished tray** flat on the table and hold it there while the housing goes in.
- With the **right gripper**, pinch the closed housing across its left and right outside walls and lift it straight up out of the nest.
- Carry it in one go to the finished tray, held level and top face up. Do not set it down on the table on the way, and do not drag it across the table.
- With the **right gripper**, lower it straight down into the **next empty slot**, filling the slots in the order 1, 2, 3, 4, with its **front mark** pointing toward the **front edge**.
- Open the **right gripper** once the housing is **seated flat** in its slot, and lift clear.
- With the **left gripper**, let go of the tray and move clear.
- Do not stack a housing on another housing, and do not put two housings in one slot.

Expected state: the nest is empty, the finished tray holds one more closed housing, and both grippers are clear.

Check: the housing sits flat in its slot, its front mark points toward the front edge, and the slots are filled in order from the front with no gaps. If it sits proud of the slot or rocks, lift it straight out with the right gripper and set it in again.

- If the finished tray holds fewer than four housings, go back to **Step 1** with the next bottom half.
- If the finished tray holds four housings, go on to **Step 7**.

Step 7: End the episode

Goal: recording ends with four closed housings in the finished tray and both arms home.

1. Confirm four closed housings sit in slots 1 to 4 of the finished tray, each seated flat with its front mark toward the front edge, and confirm the bottom tray, the top stand, and the nest are all empty.
2. Return both arms home with grippers open. Homing is the last thing the arms do.
3. Stop recording.

After the episode: reset the workspace

All reset work happens with recording off.

1. With recording off, lift each closed housing out of the finished tray and open it: press the clip hooks in through their catch windows and lift the top half straight off the bottom half. Do not pry the halves apart at the seam.
2. Stand the four bottom halves open side up in the bottom tray, each with its front mark toward the front edge, and set the tray for the next episode's config — back-left (Config L), front-center (Config M), or front-right (Config R).

3. Set the four top halves on the top stand, open side down, each with its front mark toward the front edge and its hold pad facing up.
4. Leave the assembly nest empty and all four slots of the finished tray empty.
5. Wipe dust and plastic crumbs out of the nest and out of the slots.
6. Inspect the parts. Replace a top half with a snapped, bent, or flattened clip, or with a cracked wall. Replace a bottom half with a torn or closed catch window, or a cracked wall. Replace any half that is warped so a correctly lowered top half no longer sits flush. Refit the nest if a bottom half moves in it under a gripper.
7. Run both Setup checklists before the next episode.

SOP violations

Things that break this SOP and that reviewers look for in the side-by-side review tool.

How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain multiple violations; tag each separately. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

Violations

Note on the start position: the violations below were written for Config R (bottom tray at the front-right, right gripper carries every part). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the bottom tray is not in the start zone for the config; a gripper reaches across the table for a bottom half; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the gripper on the bottom tray's side takes each bottom half and sets it in the nest; the IF line followed is the one for the

config on the table).

- **Coaching note:** look where the bottom tray is before the first reach, then follow that config's IF line through Step 1 for all four housings.

Violation: Bottom half set in the nest wrong

- **Visible cue:** the bottom half is dropped into the nest, rocks or sits high on one side, or its front mark points anywhere but the front edge, and the episode moves on.
- **SOP rule broken:** Step 1 (lower the bottom half straight down into the nest until it is seated flat, front mark toward the front edge).
- **Coaching note:** flat and marked to the front before anything goes on top of it.

Violation: More than one part carried at once

- **Visible cue:** the right gripper lifts two bottom halves, two top halves, or a half and a housing together, or holds a part while reaching for another.
- **SOP rule broken:** Steps 1, 2, and 6 (carry one part at a time, in one go, straight from its spot to the nest or the tray).
- **Coaching note:** one part per trip. A second part in the gripper is a dropped part.

Violation: Part set down on the way

- **Visible cue:** a bottom half, a top half, or a closed housing is set down on the table, on the nest rim, or on the tray edge between its spot and where it belongs.
- **SOP rule broken:** Steps 1, 2, and 6 (carry each part in one go; do not set it down on the way).
- **Coaching note:** pick it up once, put it down once.

Violation: Top half the wrong way round

- **Visible cue:** the top half is lowered on with its front mark pointing away from the front edge, or the two front marks do not line up, and the clip pass starts anyway.
- **SOP rule broken:** Step 2 (both front marks point toward the front edge before the top half is let go).
- **Coaching note:** read both marks before you let go. A housing closed backwards has to be broken open.

Violation: Top half dragged or twisted into place

- **Visible cue:** the top half lands and is then slid, dragged, or twisted on the bottom half to line the walls up, or it is landed on one corner and levered down.
- **SOP rule broken:** Step 2 (lower the top half level and straight down; if it lands crooked, lift it straight up and lower it again).
- **Coaching note:** lift and re-land it. Never shove one half across the other.

Violation: Clips pressed before the halves rest square

- **Visible cue:** a press pad is pressed while the top half still rocks, sits high on one side, or hangs over a wall of the bottom half.
- **SOP rule broken:** Steps 2 and 3 (the top half sits in rest position, level and lined up all the way round, before any pad is pressed).
- **Coaching note:** square it first. A clip pressed on a crooked half latches the housing crooked.

Violation: Halves not held together through the clip pass

- **Visible cue:** the left gripper is off the hold pad while the right gripper presses a pad, or it lets go before all four pads are pressed, and the top half lifts, rocks, or shifts.
- **SOP rule broken:** Steps 3 and 4 (the left gripper presses the hold pad and holds the halves together until all four pads are pressed and all four edges are pressed home).
- **Coaching note:** the left gripper stays on the hold pad. It is what stops the far side lifting.

Violation: Clips clicked out of order

- **Visible cue:** a higher-numbered pad is pressed while a lower-numbered pad has not been pressed, for example the right pad before the front pad, or the pads worked straight round the housing.
- **SOP rule broken:** Step 3 (press the pads in pad order 1 back, 2 front, 3 right, 4 left).
- **Coaching note:** back, front, right, left. Opposite sides in turn keep the halves parallel.

Violation: Pad pressed at an angle or rocked

- **Visible cue:** the gripper presses a pad on a tilt, rocks the top half to make a clip catch, or hooks its tip under the seam to lever the halves together.
- **SOP rule broken:** Step 3.1 (press each pad straight down; do not rock the top half, press at an angle, or pry the seam).
- **Coaching note:** straight down on the pad. A rocked clip goes in bent and comes out broken.

Violation: Housing pushed out of the nest

- **Visible cue:** the housing slides, tips, or lifts out of the nest under a press, or the gripper pushes it sideways against a nest wall.
- **SOP rule broken:** Steps 3 and 4 (press down only, and keep the housing seated in the nest through the clip pass and the edge press).
- **Coaching note:** press down, not across. If it moves, you are pushing at an angle.

Violation: Pad pressed past the click

- **Visible cue:** the gripper keeps pressing after the clip has clicked, or leans its weight on the top face after a click.
- **SOP rule broken:** Step 3.1 (stop pressing the moment the clip clicks, then lift straight up).
- **Coaching note:** one click, hands off. The click is the finish line.

Violation: Clip left unlatched

- **Visible cue:** a pad is pressed with no click and the episode moves on with no second press, or the housing is trayed with a catch window still empty.
- **SOP rule broken:** Steps 3 and 3.2 (every clip clicks once; if a pad does not click, press it once more and log the housing at Step 5).
- **Coaching note:** four windows, four hooks. Count them before you lift the housing.

Violation: Edge press skipped or short

- **Visible cue:** the housing goes to the seam check after fewer than four edges are pressed, the presses are taps with no 2-second hold, or the gripper is slid along the face instead of pressed down on each edge.
- **SOP rule broken:** Step 4 (press straight down on all four edges, back, front, right, left, holding each for 2 seconds).
- **Coaching note:** four edges, two seconds each. The clips catch; the edge press closes the seam.

Violation: Seam check skipped or partial

- **Visible cue:** the housing is lifted out of the nest with no pause to look at the seam, or only the front side is looked at before it is trayed.
- **SOP rule broken:** Step 5 (look at the seam on the front, right, back, and left before the housing is trayed).
- **Coaching note:** all four sides, every housing. A gap you did not look for still ships.

Violation: Bad seam trayed without a re-press

- **Visible cue:** daylight or a step in the seam is in plain view at the seam check, and the housing is trayed with no second press on that edge.
- **SOP rule broken:** Step 5 (press that edge again and look at that side once more before going on).
- **Coaching note:** press it again first. Only after the second press does a gap get logged.

Violation: Housing lifted before it is closed

- **Visible cue:** the housing is picked up out of the nest while a clip has not clicked or an edge has not been pressed, or the top half comes off in the gripper as it is lifted.

- **SOP rule broken:** Steps 3, 4, and 6 (all four clips click and all four edges are pressed before the housing leaves the nest).
- **Coaching note:** close it in the nest. The nest is the only thing holding the bottom half.

Violation: Housing dragged to the tray

- **Visible cue:** the closed housing is slid or dragged across the table to the finished tray, or bumped along the tray to find its slot.
- **SOP rule broken:** Step 6 (lift the housing straight up and carry it level to its slot).
- **Coaching note:** lift it clear and carry it. Dragging pops clips open.

Violation: Finished tray not steadied

- **Visible cue:** the left gripper is off the finished tray while a housing goes in, or the tray slides, tips, or lifts as the housing is lowered.
- **SOP rule broken:** Step 6 (the left gripper holds the tray flat on the table while the housing goes into its slot; the tray is never lifted).
- **Coaching note:** pin the tray first, then lower the housing.

Violation: Trayed in the wrong place or the wrong way

- **Visible cue:** a slot is skipped, a housing goes into a slot out of order, two housings share a slot, a housing is stacked on another one, it sits proud or rocking, or its front mark points away from the front edge.
- **SOP rule broken:** Step 6 (fill the slots in the order 1, 2, 3, 4, one housing per slot, seated flat with the front mark toward the front edge).
- **Coaching note:** front slot first, then back one at a time. One housing, one slot.

Violation: Two housings worked at once

- **Visible cue:** a second bottom half is picked up before the housing in the nest is trayed, or a part is staged on the table beside the nest to be used next.
- **SOP rule broken:** Steps 1–6 (finish and tray one housing before picking up the next bottom half; only one housing is in the nest at a time).
- **Coaching note:** one housing at a time, start to tray.

Violation: Work done out of order

- **Visible cue:** the steps run out of sequence, for example the top half goes on before the bottom half is seated flat, an edge is pressed before all four clips have clicked, or the seam is checked before the edge press.
- **SOP rule broken:** Steps 1–6 (bottom half in, top half on, four clips, four edges, seam check, tray, in that order).
- **Coaching note:** the order is the task. Say the next step's name before you move.

Violation: Fewer than four housings built

- **Visible cue:** recording stops with fewer than four closed housings in the finished tray, or with a bottom half or top half left in the bottom tray or on the top stand.
- **SOP rule broken:** Steps 1–7 (all four housings are built and trayed in the one episode before it ends).
- **Coaching note:** the episode is four housings. Count the slots before you home the arms.

Violation: Wrong arm used

- **Visible cue:** an action assigned to one gripper is done by the other, including the left gripper carrying a half, pressing a press pad, pressing an edge, or traying a housing; or the right gripper holding the hold pad or steadying the finished tray.
- **SOP rule broken:** Steps 1–6 (the right gripper carries the parts, clicks the clips, presses the edges, and trays the housing; the left gripper holds the hold pad and steadies the tray).
- **Coaching note:** right gripper does the work, left gripper holds and steadies.

Violation: Part or tray knocked off its spot

- **Visible cue:** a half or a closed housing is dropped on the table, into the nest, or on the floor; the bottom tray, the top stand, or the finished tray is tipped, spilled, or pushed off its spot; or the nest is knocked out of square.
- **SOP rule broken:** Steps 1–6 (keep the parts, the trays, the stand, and the nest on their spots through the episode).
- **Coaching note:** work slower and lower over the table, and keep every carry short.

Violation: Wrong episode ending

- **Visible cue:** recording stops before the end state is confirmed, a part is left in the nest or on a supply spot, an arm is not home, a gripper is closed, or an arm does something else after homing.
- **SOP rule broken:** Step 7 (confirm four trayed housings and empty supply spots, return both arms home with grippers open as their final action, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

Failures not caused by how the task was run are system issues. Log and discard the episode rather than tagging them as SOP violations.

- Recording stops or pauses during the episode.
- A camera drops frames or loses its feed.
- An arm or gripper fails, drifts, or reports a motor error.
- A clip is snapped off, bent flat, or moulded short, so a correctly pressed pad never clicks.

- A catch window is moulded closed or torn open, so a correctly pressed clip cannot latch.
- A half is warped or bent, so a correctly lowered top half cannot sit flush on the bottom half.
- A half cracks or splits under a correct straight-down press.
- The assembly nest is loose or the wrong size, so the bottom half moves however carefully a gripper presses.
- A finished tray slot is the wrong size, so a correctly closed housing will not sit flat in it.

Annotation subtasks (from SOP)

1. Carry a bottom half from the bottom tray into the assembly nest
2. Carry a top half from the top stand and lower it onto the bottom half
3. Hold the halves together with the left gripper
4. Press one press pad to a click
5. Press one edge of the top face home
6. Look at the seam on all four sides
7. Lift the closed housing out of the nest
8. Set the closed housing into its slot in the finished tray
9. Confirm the end state, return both arms home, and end the episode